Objective-Aware Projection Policies for Inverse PDE Posterior Sampling

A Two-Axis Study of Schedule and Order under Partial Observations

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Abstract

This report studies inverse PDE posterior sampling under sparse observations, a setting common in data assimilation and scientific sensing where uncertainty, physical validity, and runtime all matter. DiffusionPDE introduced generative PDE solving under partial observation, and Physics-Constrained Flow Matching (PCFM) introduced projection-based constraint correction during sampling. Building on these ideas, I examine which projection schedule (when to project) and projection order (how strongly to project) best trade off inverse reconstruction quality, physical consistency, runtime, and trajectory stability. The nonlinear elliptic benchmark

$$-\Delta v + \kappa v^3 = u, \quad \kappa = 50, \quad N_x = 63,$$

is used with periodic boundary conditions, where full u is recovered from sparse v observations. Relative to baseline, the best tuned setting improves u_error by **21.75%** ($1.9232 \rightarrow 1.5050$) with a clear quality-runtime tradeoff. I then extend the same protocol across additional PDE families, where guidance gains range from -9.03% (Burgers BC) to -29.84% (reaction-diffusion). Across both studies, projection policy is objective-aware rather than globally fixed. More broadly, the results show that schedule/order is a practical control knob for flow-matching and other constrained generative samplers, because correction effort can be allocated across the trajectory to match application priorities.

1 Concepts Used

MEAM 4600 concepts used include PDE modeling, inverse formulation under partial observations, residual-based numerical correction, and objective-wise tradeoff analysis. I use these concepts to structure model design, inverse inference, and objective-wise evaluation.

Core formulas:

- PDE model: $-\Delta v + 50v^3 = u$ with periodic BC.
- Inverse target: recover full u from partial v observations.
- Residual: $r(u,v) = -\Delta v + 50v^3 - u$.
- Sampling + guidance:

$$x_{k+1} = x_k + \Delta t\, f_\theta(x_k, t_k) - g\nabla_x \mathcal{L}_{\text{obs}}(x_k).$$

- Projection:

$$x_{k+1} \leftarrow x_{k+1} - P_q^{-1}(x_{k+1})\, r(x_{k+1}),$$

  with first-order and Gauss-Newton variants.

2 Problem Framing

In inverse PDE settings, sparse measurements make recovery inherently ambiguous: many full fields can match the same observed subset. Posterior sampling is therefore used instead of a single deterministic predictor, so uncertainty can be represented and then guided toward physically plausible solutions.

Given that context, the research question is: *which projection schedule and projection order are best for each target objective?* The focus is on mechanism and tradeoffs, so the goal is principled comparison rather than state-of-the-art claims.

3 Method Summary

The benchmark studied is

$$-\Delta v + 50v^3 = u, \qquad N_x = 63, \qquad \text{periodic BC.}$$

Only 10% of v is observed, and full u is recovered. This observation mask is kept fixed across comparisons so schedule/order remains the main changing factor; I keep this fixed-mask protocol to isolate schedule/order effects cleanly.

The joint state is represented as $x = [u; v] \in \mathbb{R}^{2N_x}$, with residual

$$r(x) = -\Delta v + 50v^3 - u.$$

At each sampling step, the state first moves along the learned flow with observation guidance:

$$x_{k+1} = x_k + \Delta t\, f_\theta(x_k, t_k) - g\, \nabla_x \mathcal{L}_{obs}(x_k),$$

then optional projection is applied:

$$x_{k+1} \leftarrow x_{k+1} - P_q^{-1}(x_{k+1})\, r(x_{k+1}),$$

where schedule S controls *when* projection is applied, and order $q \in \{\texttt{first\_order}, \texttt{gauss\_newton}\}$ controls *how* each correction is computed. These two axes are separated because timing and correction strength affect different failure modes (instability, over-correction, or weak physics enforcement).

The method is intentionally simplified: one prior family, one inverse formulation, and controlled schedule/order/guidance sweeps. This scope keeps conclusions interpretable and reproducible.

> **Baseline definition.** Baseline means the default flow-matching model trained for 80 epochs at default width/depth, evaluated on the same projection grid. This keeps the comparison fair: gains come from tuning and projection policy, not from framework changes.

4 Metrics and Protocol

To make the evaluation logic explicit before presenting numbers, metric roles are first mapped to their exact function in the inverse task.

Table 1: Metric roles in this inverse setting.

Metric	**Interpretation**
`u_error`	Primary inverse target quality
`v_error`	Full-state reconstruction, incl. unobserved regions
`obs_error`	Fit on observed entries only
`posterior_quality`	Composite ranking score
`physical_consistency`	Pre-cleanup PDE consistency
`runtime` / `trajectory_stability`	Efficiency / robustness
`mmse, smse, fpd`	Distribution-level comparison metrics
`ce_ic, ce_bc, ce_cl`	Constraint-specific errors (N/A when inapplicable)

Evaluation uses 128 samples × 3 observation seeds in the nonlinear main study. Observation-guidance strength is swept up to 16, then baseline is compared with a stronger checkpoint (`e200_big`: 200 epochs, larger model width/depth) under identical settings. This isolates whether gains come from training, guidance strength, or projection policy.

5 Results: Nonlinear Elliptic

5.1 Objective-wise winners

The table below gives the best projection policy per objective under one fixed evaluation setup.

Table 2: Best schedule/order by objective.

Objective	**Winning run**	**Best policy**	**Value**
`posterior_quality`	larger 200-epoch checkpoint + strong guidance	project every 5 steps / Gauss-Newton correction	3.1364
`physical_consistency`	larger 200-epoch checkpoint + strong guidance	project every step / Gauss-Newton correction	0.000359
`runtime`	baseline (default 80-epoch checkpoint)	no projection	0.6086 s
`trajectory_stability`	baseline (default 80-epoch checkpoint)	no projection	0.0

These winners are numerically distinct: best `posterior_quality` is 3.1364 with every_5/Gauss-Newton, best `physical_consistency` is 0.000359 with every_step/Gauss-Newton, while the fastest and most stable policy is no projection (0.6086 s, trajectory_stability = 0.0). One global policy is therefore insufficient. In plain words: the stronger checkpoint is a larger model trained for 200 epochs; strong guidance means a high weight on matching observations; "project every step" and "project every 5 steps" define correction frequency; first-order and Gauss-Newton are two correction rules; "no projection" means no correction pass.

> **Key result.** Compared with the baseline best row ($u\_error = 1.9232$, $v\_error = 1.2944$, $obs\_error = 1.1504$, $post\_q = 4.3679$, runtime = 0.7357 s), the tuned setup (strong guidance + first-order projection every step) reaches $u\_error = 1.5050$, $v\_error = 0.9120$, $obs\_error = 0.8126$, $post\_q = 3.2295$, runtime = 1.3741 s. This corresponds to −21.75%, −29.54%, −29.36%, and −26.06% improvements, with a +86.77% runtime cost.

5.2 Best inverse row vs baseline

To evaluate inverse recovery directly, the best-u row from baseline is compared against the best-u row after tuning. Best inverse rows are compared directly because u_error is the primary metric for the inverse target. Row-to-row, u_error drops by 0.4182 ($1.9232 \to 1.5050$), v_error by 0.3824 ($1.2944 \to 0.9120$), obs_error by 0.3378 ($1.1504 \to 0.8126$), and $post\_q$ by 1.1384 ($4.3679 \to 3.2295$), while runtime increases by 0.6384 s ($0.7357 \to 1.3741$). The best inverse-reconstruction setup is strong guidance with first-order projection at every step, possibly because frequent lightweight corrections keep each sample close to the observation-consistent region while avoiding aggressive per-step overcorrection.

> **Key nonlinear conclusion.** If the goal is best inverse reconstruction, project at every step with first-order correction ($u\_error = 1.5050$, $v\_error = 0.9120$). If the goal is best physics consistency, use Gauss-Newton-heavy correction, because curvature-aware updates reduce nonlinear PDE residual more strongly per correction; in this study, that policy achieved $physical\_consistency = 0.000359$. If the goal is speed or maximum stability, skip projection (runtime = 0.6086 s, trajectory_stability = 0.0).

6 Results: Multi-Family Extension

To test whether the nonlinear finding is specific to one PDE, the same pipeline is applied to linear elliptic, heat, reaction-diffusion, Burgers (IC/BC), and a 1D Navier-Stokes surrogate. The same evaluation logic is kept so cross-family conclusions remain comparable.

The following table reports how much inverse-target error drops as guidance strength increases from weak to strong.

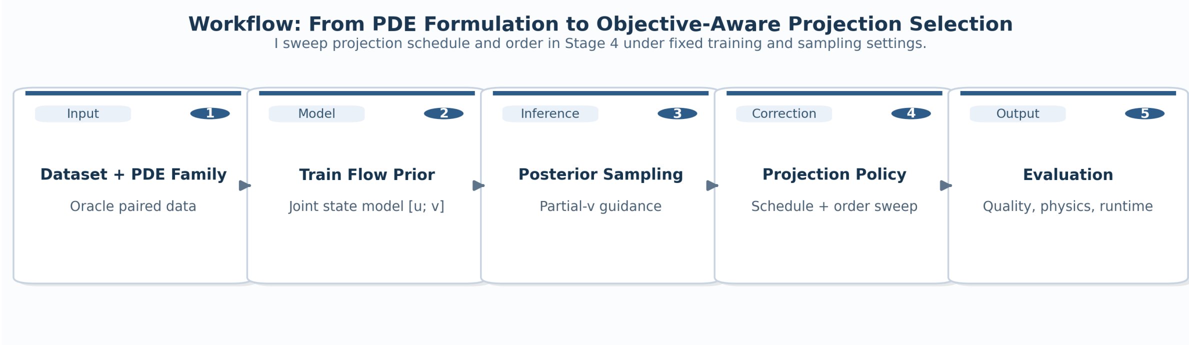

Figure 1: Workflow from PDE-family data to objective-wise schedule/order ranking. Projection is inserted inside the sampling trajectory and swept systematically.

Table 3: Best inverse row comparison (128×3).

Setting	u_err	v_err	obs_err	post_q	runtime
Baseline best row	1.9232	1.2944	1.1504	4.3679	0.7357 s
Best tuned (strong guidance, projection every step, first-order correction)	1.5050	0.9120	0.8126	3.2295	1.3741 s
Relative change	−21.75%	−29.54%	−29.36%	−26.06%	+86.77%

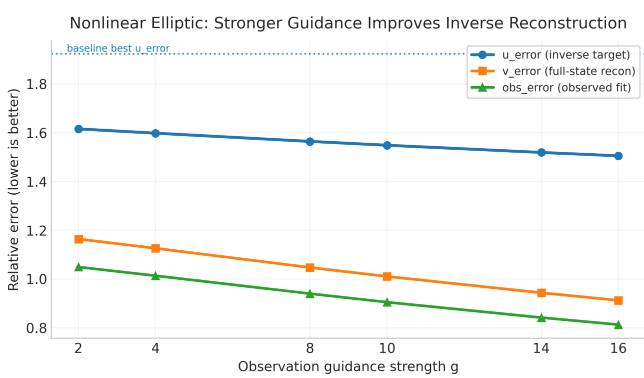

Figure 2: Stronger guidance consistently improves reconstruction metrics.

Table 4: Best-row u_error gain from $g = 0.25$ to $g = 16$.

Family	u_error reduction
Linear elliptic	−18.73%
Heat equation	−25.20%
Reaction-diffusion	−29.84%
Burgers IC	−26.12%
Burgers BC	−9.03%
Navier-Stokes 1D surrogate	−24.11%

Across families, stronger observation guidance consistently improves inverse recovery, with u_error gains of −18.73%, −25.20%, −29.84%, −26.12%, −9.03%, and −24.11% for linear elliptic, heat, reaction-diffusion, Burgers IC, Burgers BC, and Navier-Stokes surrogate, respectively. Exact schedule/order winners remain objective-specific, matching the nonlinear benchmark.

7 Reflection and Reproducibility

What worked. Nonlinear inverse quality improved materially ($u\_error : 1.9232 \to 1.5050$, $v\_error : 1.2944 \to 0.9120$), and complete objective-wise ranking tables were produced.

What did not always work. Better reconstruction required higher compute (runtime $0.7357 \to 1.3741$ s), and larger/longer training was not universally better across families.

What I learned. Projection policy should be selected by objective: every-step first-order gave best inverse recovery, every-step Gauss-Newton gave best physics consistency (0.000359), and no projection gave best runtime/stability.

Attribution and responsibility. Existing repository components (especially `src/chonkdiff`) were used as a numerical backend, and the posterior-projection study was implemented/extended in `src/posterior_projection`. LLM assistance was used for coding/report drafting, and metric values, commands, and final conclusions were manually verified before submission.

8 Limitations and Next Steps

The method was intentionally simplified, which improved interpretability but limits direct comparability to larger benchmark suites. In particular, full-scale reproduction of all external baselines and exhaustive architecture searches were not attempted. A natural next step is to reuse this same evaluation protocol in broader cross-method studies (using `mmse/smse/fpd` and family-specific constraint errors) while preserving the objective-wise projection analysis developed here.

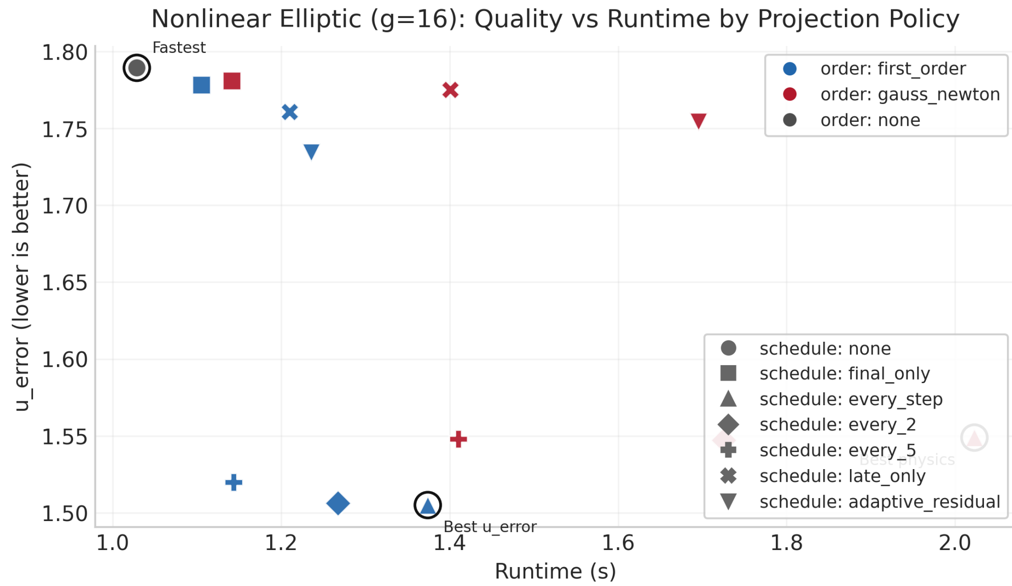

Figure 3: Quality-runtime frontier at guidance strength 16. Color shows correction rule, marker shows projection schedule, and rings mark best inverse quality, fastest runtime, and best physics consistency.

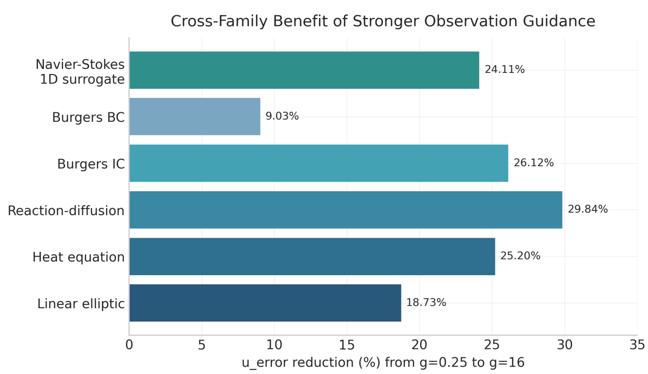

Figure 4: Cross-family trend of stronger-guidance benefit.

9 Conclusion

Projection schedule/order is a first-class algorithmic choice in partial-observation inverse PDE sampling. On the proposal benchmark, the strongest inverse recovery is achieved by strong guidance plus every-step first-order projection ($u\_error = 1.5050$, $v\_error = 0.9120$); best physics consistency is achieved by every-step Gauss-Newton projection (0.000359); and best runtime/stability is achieved by no projection (0.6086 s, 0.0). Therefore, projection design should match the objective of interest rather than using one fixed policy.

References

[1] J. Huang, G. Yang, Z. Wang, and J. J. Park, *DiffusionPDE: Generative PDE-Solving Under Partial Observation*, 2024.

[2] U. Utkarsh, P. Cai, A. Edelman, R. Gomez-Bombarelli, and C. V. Rackauckas, *Physics-Constrained Flow Matching: Sampling Generative Models with Hard Constraints*, 2025.

[3] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, *Flow Matching for Generative Modeling*, 2022.

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